Complex Dynamics of Multichannel Reinforcement Learning: Single Agent Case

Ihor LUBASHEVSKY[1], Arkady ZGONNIKOV[2],
Alieksiei KOROTKOV[3], Namik GOUSSEIN-ZADE[4]

[1,2]University of Aizu, Tsuruga, Ikki-machi, Aizu-Wakamatsu, Fukushima, 965-8580 Japan
[3,4]General Physics Institute, Russian Academy of Sciences, Vavilov str., 38, Moscow, 119991 Russia

E-mail: [1]i-lubash@u-aizu.ac.jp, [2]arkady.zgonnikov@gmail.com, [3]alieksiei_korotkov@inbox.ru, [4]ngus@mail.ru

Abstract

We develop an adaptation model based on the reinforcement learning framework extended by an additional processing channel. We scrutiny the dynamics of the single agent adapting to the unknown environment; the agent is biased by novelty seeking, the intrinsic inclination for exploration. We demonstrate that the behavior of the novelty-seeking agent may be extremely irregular and the concepts of chaos and intermittency may be used to characterize such reinforcement learning. Our results give evidence to the hypothesis that the intrinsic motives of agents should be paid no less attention than the extrinsic ones in the models of complex socio-economic systems.

Keyword: Reinforcement Learning, Intrinsic Motivation, Novelty, Complex Dynamics, Emergence

Model

Learning and adaptation are among the basic mechanisms of emergent phenomena in many socio-economic systems. A fundamental feature of human beings is the trait of novelty seeking [1] and in the context of learning it propels humans to actively learn and explore [2, 3].

Recently [4] we developed a dynamical model of single-agent adaptation as a richer alternative to standard reinforcement learning. In the simplest situation, a single agent faces an unknown environment, which is represented by a number of rewarded actions. The agent gradually learns the initially unknown rewards by making a repeated choice between these actions. Canonically, the ultimate goal of the agent is to maximize the total sum of the rewards gained throughout the process. In constructing the model we introduced the additional channel of information processing to reinforcement learning (Fig. 1). We then analyzed in detail how the adaptation dynamics changes when the agent behavior is governed not only by the external factors (that is, rewards), but also by the agent intrinsic motives, namely, the inclination to engage in novel activities. As shown, the behavior of the novelty-seeking agent may be inherently unstable.

In the present work we demonstrate that the individual behavior of a single novelty-seeking agent can become complex when the set of possible alternatives of its repeated choice contains more than two elements. Moreover, when the number of possible alternatives exceeds three the behavior of this agent can be highly irregular.

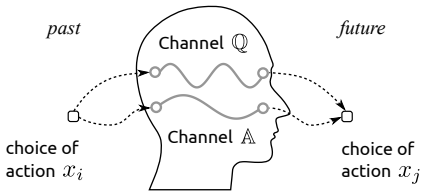


Figure1: Two channels of information processing. Channel ℚ which processes the information about the rewards continuously interacts with channel 𝔸 which deals with all the information not directly related to the rewards.

In the continuous approximation the analyzed novelty-seeking agent adaptation is described by the following equations:

$$\dot{q}_i = r_i p_i^{1-\gamma} - \frac{q_i}{T_q}, \quad \dot{a}_i = \phi_i p_i - \frac{a_i}{T_a},$$
$$p_i = \frac{e^{q_i - a_i}}{\sum_j e^{q_j - a_j}}. \qquad (1)$$

Here i is the index of one of N alternatives for the agent choice, q_i and a_i are the variables quantifying the "quality" of alternative i via the channels ℚ and 𝔸, respectively, r_i and ϕ_i are the reward and the "effective" reward gained via the two channels, T_q and T_a are the time scales of the information loss via these channels, and p_i is the current probability of choosing alternative i affected by the two

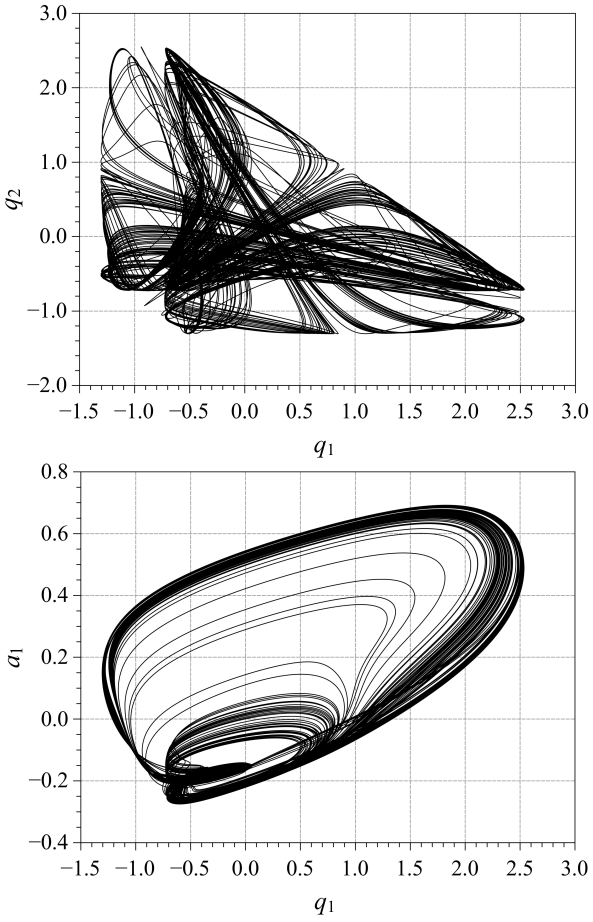


Figure2: The phase portrait of the long-term transient process found in the system dynamics in the case of four alternatives, $N = 4$, of the agent choice. The given figure shows the projections of the system trajectories in the multi-dimensional phase space onto the planes $\{q_1, q_2\}$ and $\{q_1, a_1\}$.

channels simultaneously.

Numerical Results

System (1) was studied numerically. Figure 2 demonstrates the fact that the given system does exhibit complex behavior. When the number of the choice alternatives is four or larger, $N \geq 4$, there can appear an extremely long and complex transient process before the system gets one of the stable limits cycles. For the case shown in Fig. 2 the transient process can comprise up to 200 "elementary" oscillations. As seen, the system dynamics during this transient process has to be categorized as highly irregular and the terms chaos and intermittency are the relevant notions to describe it.

The found results also enable us to pose the following two related questions for discussion. First, we hypothesize the existence of so long-term transient process to be due to the close location of different stable limit cycles, at least, some of their parts, in the multi-dimensional phase space. So we may expect that rather small noise must convert this transient process into a stable state of the agent choice dynamics.

The second rather general question concerns the concept of noise-induced chaos as well as noise-induced intermittency discussed in literature (see, e.g., [5, 6] and references therein). In the analyzed model the system cannot "find" an attractor in the phase space for a long time during which its dynamics looks like chaos with pronounce intemittency. It is due to the nonlinear properties of the regular (deterministic) "forces" acting on the system. A possible random factor just prevents the system from getting an attractor or forces it to escape from the attractor. So the local characteristics of such irregular motion have to be specified by the regular "forces" on their own. The analysis of various particular systems admitting this type chaoticity should enable us to penetrate deeper onto understanding these emergent phenomena seeming to be rather typical for human society.

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